

Readiness and Responsible AI Awareness: A UTAUT-Based Study of Behavioral Intention among Final-Year Students and Young Professionals in Ho Chi Minh City

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Conference track: Track 4 — Governance, Policy and Societal Transformation

Total word count: 7,378

Note: Author information appears only on this cover page and is omitted from the main manuscript to preserve the integrity of the double-blind peer-review process.

Abstract

This study examines the factors influencing behavioral intention to adopt artificial intelligence (AI) responsibly among final-year university students and young professionals (< 2 years of work experience) in Ho Chi Minh City, Vietnam. Building on the Unified Theory of Acceptance and Use of Technology (UTAUT), the study integrates four core predictors — Effort Expectancy, Performance Expectancy, Social Influence, and Facilitating Conditions — with two context-specific moderating constructs, Responsible AI Awareness and AI Anxiety, to address a gap in the Vietnamese literature regarding responsible AI use intention among labour-market entrants. Data were collected from 236 valid respondents via an online survey and analyzed using confirmatory factor analysis and covariance-based structural equation modeling (CB-SEM) with bootstrapping (5,000 resamples). Of six hypothesized paths, two were statistically supported: Social Influence exerted the strongest effect on behavioral intention ($\beta = 0.379$, $p < .001$), followed by Effort Expectancy ($\beta = 0.211$, $p = .044$). Performance Expectancy and Facilitating Conditions were non-significant, and neither Responsible AI Awareness nor AI Anxiety significantly moderated the UTAUT - intention relationships. The model explained 45.4% of variance in behavioral intention. Findings suggest that for this digital-native population, responsible AI adoption is driven primarily by social-normative influence rather than utilitarian calculation, with implications for universities, employers, and policymakers seeking to promote responsible AI use.

Keywords: Responsible AI; AI literacy; UTAUT; behavioral intention; AI anxiety; young workforce; Ho Chi Minh City

1. Introduction

The rapid advancement of artificial intelligence (AI), and particularly generative AI (GenAI), is reshaping how individuals learn, work, and access information. AI is increasingly embedded in information retrieval, data analysis, content generation, task automation, decision support, and customer interaction, raising new expectations for the competencies and attitudes required of AI users - especially students and young professionals who are preparing to enter, or have recently entered, the labour market. According to the World Economic Forum (2025), structural change in the labour market between 2025 and 2030 is projected to affect approximately 22% of current jobs, with around 170 million new roles created and 92 million existing roles displaced, yielding a net gain of roughly 78 million jobs; an estimated 39% of workers' existing skills are expected to change or become obsolete over the same period, with AI - and data-related skills rising sharply in demand.

Within this context, the capacity to use AI has become an increasingly important professional competency. However, being able to operate an AI tool is not equivalent to being ready to use AI appropriately and responsibly. Users must be able to recognize AI's limitations, judge the reliability of AI-generated content, identify risks related to data, privacy, and bias, and maintain human oversight and accountability. UNESCO's (2024) AI competency framework emphasizes that learners' AI competency should be built around human-centred thinking, AI ethics, AI techniques and applications, and AI system design — foregrounding safe, meaningful, and responsible engagement with AI rather than mere tool proficiency.

The Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al., 2003) offers a well-established framework for explaining technology acceptance through Effort Expectancy, Performance Expectancy, Social Influence, and Facilitating Conditions, and has recently been extended to AI contexts. Ke et al. (2025) combined UTAUT with AI literacy to

explain AI adoption intention among Chinese university students, while in Vietnam, Nghĩa (2026) integrated TAM and UTAUT - adding AI anxiety - to explain GenAI usage intention, finding significant effects of Performance Expectancy, Social Influence, Effort Expectancy, and Facilitating Conditions on Behavioral Intention. In Ho Chi Minh City specifically, Chung et al. (2024) examined factors influencing university students' use of AI tools for learning, and Vo and Nguyen (2024) examined English-major students' acceptance of ChatGPT for language learning. These studies confirm the relevance of UTAUT in the Vietnamese context but remain concentrated on AI use for academic purposes, leaving the transition into professional, work-related AI use largely unexamined.

More importantly, intention to use AI is not equivalent to intention to use AI *responsibly*. An individual may hold a strong intention to use AI because it is perceived as useful and easy to use, while remaining insufficiently aware of risks related to data, bias, misinformation, or accountability for AI-assisted outputs. Concepts of AI literacy (Wang et al., 2022) and Responsible AI (UNESCO, 2021; NIST, 2023) suggest that responsible use requires awareness of transparency, fairness, data protection, bias, verifiability, accountability, and human oversight - dimensions not captured by conventional technology-acceptance constructs alone. At the same time, AI anxiety - apprehension, discomfort, or unease toward AI arising from uncertainty about its capabilities, fear of losing control, or concern about job displacement - has been shown to shape technology engagement independently of utilitarian beliefs (Chen et al., 2024; Cao et al., 2026).

In Vietnam, workforce readiness for AI has also been identified as a national policy priority. Prime Ministerial Decision No. 127/QĐ-TTg (26 January 2021), promulgating the National Strategy on Research, Development, and Application of Artificial Intelligence to 2030,

designates AI human-resource development as a core pillar for building Vietnam's AI research, development, and application capacity. Ho Chi Minh City, as the country's largest concentration of universities, enterprises, and digital-economy activity, provides an appropriate setting in which to examine this readiness.

Despite this growing body of work, two gaps remain. **First**, to the authors' knowledge, existing research in Vietnam has paid limited attention to Responsible AI Awareness as a construct influencing behavioral intention among young labour-market entrants, as distinct from general AI-use intention. **Second**, final-year students and recent graduates (< 2 years of work experience) — the group at the direct threshold of the school-to-work transition — have not been studied as a distinct population, despite their AI readiness having direct implications for workforce digital capability. This study addresses these gaps by integrating four core UTAUT predictors (Effort Expectancy, Performance Expectancy, Social Influence, Facilitating Conditions) with two moderating constructs — Responsible AI Awareness and AI Anxiety — to explain behavioral intention to use AI responsibly among this population in Ho Chi Minh City.

Accordingly, this study seeks to answer the following research questions: (1) How do Effort Expectancy, Performance Expectancy, Social Influence, and Facilitating Conditions affect behavioral intention to use AI responsibly among students and young professionals in Ho Chi Minh City? (2) Does Responsible AI Awareness moderate the relationships between these UTAUT predictors and behavioral intention? (3) Does AI Anxiety moderate these same relationships? In addressing these questions, the study makes two contributions. Theoretically, it extends UTAUT beyond general AI-use intention into the specific normative domain of responsible AI use, and tests, for the first time in the Vietnamese context, the joint moderating roles of AI-related awareness and AI-related anxiety. Practically, the findings offer evidence to

inform how universities, employers, and training institutions might design interventions to strengthen AI competency, responsible-use awareness, and psychological readiness among students and young professionals entering an AI-integrated workforce.

2. Literature Review and Hypothesis Development

2.1 The UTAUT and UTAUT2 Frameworks

The Unified Theory of Acceptance and Use of Technology (UTAUT), developed by Venkatesh, Morris, Davis, and Davis (2003), synthesizes eight prior technology-acceptance models — including the Theory of Reasoned Action, the Technology Acceptance Model, the Theory of Planned Behavior, the Innovation Diffusion Theory, and Social Cognitive Theory — into a single integrated framework capable of explaining up to 70% of the variance in technology usage intention. UTAUT identifies four core determinants of behavioral intention: Performance Expectancy, defined as "the degree to which an individual believes that using the system will help him or her to attain gains in job performance" (Venkatesh et al., 2003, p. 447); Effort Expectancy, "the degree of ease associated with the use of the system" (Venkatesh et al., 2003, p. 450); Social Influence, the degree to which an individual perceives that important others believe he or she should use the technology; and Facilitating Conditions, the degree to which an individual believes that organizational and technical infrastructure exists to support use.

Venkatesh, Thong, and Xu (2012) extended the original model into UTAUT2, adapted to voluntary consumer-technology contexts, adding Hedonic Motivation, Price Value, and Habit, and improving explained variance in behavioral intention from 56% to 74%. Although UTAUT2's consumer-context extensions are less directly applicable to a mandatory or semi-mandatory professional setting, its underlying logic — that technology acceptance is shaped jointly by individual cognition and environmental facilitation — remains foundational to

the present study, which retains the four original UTAUT predictors as the core explanatory mechanism.

Recent research has applied UTAUT and its extensions specifically to AI adoption. Kavitha and Joshith (2024) extended UTAUT2 with trustworthiness, personal innovativeness, perceived task excellence, and perceived privacy concern to study Gen Z AI-tool adoption intention among Indian university students ($N = 430$, CB-SEM via AMOS), explaining 34.2% of variance in intention. Ke, Gong, and Ke (2025) combined AI literacy with the four core UTAUT constructs to explain AI adoption among Chinese university students, demonstrating that pairing technology-acceptance predictors with AI-specific competency constructs yields a more comprehensive account of AI adoption intention than either approach alone — an architecture closely mirrored in the present study's design. In the Vietnamese context, Nghĩa (2026) integrated TAM and UTAUT, adding AI anxiety, to study GenAI usage intention among Vietnamese university students, finding significant effects of Performance Expectancy, Social Influence, Perceived Enjoyment, Effort Expectancy, and Facilitating Conditions on Behavioral Intention — providing direct empirical grounding for applying UTAUT constructs to this study's Vietnamese sample.

2.2 Responsible AI Awareness

As AI systems become more autonomous and consequential, technology-acceptance research increasingly recognizes that behavioral intention alone does not capture whether use is appropriate or safe. Dignum (2020, p. 4) argues that "responsible AI implies the need for mechanisms that enable AI systems to act according to ethics and human values," while Dafoe (2018) frames AI governance as concerned with "the institutions and contexts in which AI is built and used." The OECD AI Principles articulate concrete normative pillars for responsible AI

— non-discrimination and fairness, transparency, and accountability, under which "AI actors should be accountable for the proper functioning of AI systems." UNESCO (2021) similarly holds that ethical AI governance must be grounded in human rights, transparency, fairness, data protection, and human oversight, and — critically — that the use of an AI system does not diminish humans' ultimate responsibility for AI-related decisions and actions. The NIST AI Risk Management Framework (2023) reinforces this view from a risk-governance perspective, emphasizing that AI risk must be managed throughout the design, development, deployment, use, and evaluation of a system, requiring users to recognize risk, verify outputs, and maintain human control.

These principles converge with the concept of AI literacy, defined by Long and Magerko (2020, p. 2) as "a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace." Wang, Rau, and Yuan (2022) operationalized this construct through a four-dimension scale — Awareness, Use, Evaluation, and Ethics — demonstrating that AI competency is inherently multidimensional rather than purely technical. Passlack, Hammerschmidt, and Posegga (2025) extend this reasoning into professional settings, arguing that as AI systems grow more autonomous, AI literacy for knowledge workers must extend beyond productivity gains to include recognition of unintended consequences, risk, and accountability in human–AI interaction — a perspective directly relevant to students and young professionals at the threshold of the school-to-work transition, where AI use increasingly intersects with organizational data, professional decisions, and accountability for outcomes.

Building on these foundations, this study defines Responsible AI Awareness (RA) as an individual's degree of awareness of the principles and requirements that should be considered

when using AI, encompassing transparency, fairness, data protection and privacy, recognition of bias and error, verifiability of outputs, accountability, and the role of human oversight. Rather than treating RA as a standalone competency variable, this study positions RA as a moderator of the relationship between UTAUT predictors and behavioral intention, testing whether individuals with higher awareness of responsible-AI principles translate the same levels of perceived usefulness, ease of use, social encouragement, or facilitating conditions into behavioral intention differently than individuals with lower awareness.

2.3 AI Anxiety

AI anxiety refers to an individual's state of apprehension, unease, or discomfort when encountering, using, or contemplating AI. Distinct from cognitive risk awareness, AI anxiety reflects an affective, psychological response: an individual may be well aware of AI's risks without feeling anxious about them, or may experience high anxiety despite limited technical knowledge of AI. Chen, Liu, and Liu (2024), studying university students using AI for second-language learning, found that AI self-efficacy was negatively associated with AI anxiety, while AI anxiety in turn exerted a negative effect on AI use — evidence that affective responses to AI can meaningfully constrain technology engagement independent of utilitarian beliefs. Cao et al. (2026), extending an integrated TAM–UTAUT model, similarly examined the relationship between AI anxiety and GenAI usage intention in higher education, reinforcing that psychological responses to AI can shape acceptance processes alongside conventional usefulness- and ease-of-use-based predictors. In the Vietnamese context, Nghĩa (2026) incorporated AI anxiety directly into a UTAUT-based model of GenAI adoption intention among university students, lending further support to its relevance for this study's population.

For students and young professionals — a group for whom AI is simultaneously viewed as a productivity-enhancing tool and a source of uncertainty about future skill requirements and career prospects — an individual may perceive AI as both useful and easy to use while still harbouring considerable anxiety about dependency, job displacement, misinformation, or loss of control over outcomes. This study therefore positions AI Anxiety (ANX) as a second moderator, testing whether the strength of the relationship between UTAUT predictors and behavioral intention is attenuated among individuals reporting higher anxiety toward AI.

2.4 Hypothesis Development

Effort Expectancy reflects the perceived ease of learning and using AI tools. Ke et al. (2025) and Nghĩa (2026) both confirm Effort Expectancy as a significant predictor of AI-related behavioral intention, and Alotaibi, Sonbul, and El-Dakhs (2025) similarly find perceived ease of use significant for ChatGPT acceptance even as users transition from traditional to generative AI tools. Accordingly:

H1: Effort Expectancy has a positive effect on Behavioral Intention to use AI responsibly.

Performance Expectancy has consistently been shown to be among the strongest predictors of technology-acceptance intention across UTAUT-based studies (Venkatesh et al., 2003, 2012), and recent AI-specific applications confirm its continued relevance: Nghĩa (2026) found a significant effect of Performance Expectancy on Behavioral Intention toward GenAI among Vietnamese students, and Alotaibi et al. (2025), using TAM, found perceived usefulness to be a significant driver of ChatGPT acceptance. Accordingly:

H2: Performance Expectancy has a positive effect on Behavioral Intention to use AI responsibly.

Social Influence captures the extent to which individuals perceive that important others — peers, instructors, or supervisors — encourage AI use. Nghĩa (2026) found Social Influence to be a

significant predictor of GenAI usage intention among Vietnamese students, consistent with the broader UTAUT literature's emphasis on normative pressure in collectivist contexts such as Vietnam. Accordingly:

H3: Social Influence has a positive effect on Behavioral Intention to use AI responsibly.

Facilitating Conditions reflect the perceived availability of resources, knowledge, devices, and support needed to use AI. Although this construct is often significant in early-stage technology adoption, its effect may attenuate once infrastructural access becomes widespread — as is increasingly the case for free, browser-based generative AI tools. Nonetheless, consistent with the original UTAUT formulation and its confirmed relevance in Nghĩa's (2026) Vietnamese sample, this study hypothesizes:

H4: Facilitating Conditions has a positive effect on Behavioral Intention to use AI responsibly.

Responsible AI Awareness, grounded in the ethical and governance principles articulated by UNESCO (2021), NIST (2023), and the OECD AI Principles, is expected to condition how utilitarian and social cognitions translate into behavioral intention. An individual with higher awareness of AI's risks, transparency requirements, and accountability obligations may evaluate the same level of perceived usefulness, ease of use, social encouragement, or facilitating conditions more deliberately, altering the strength of these relationships relative to individuals with lower awareness (Passlack et al., 2025). Accordingly:

H5: Responsible AI Awareness moderates the relationship between overall UTAUT-related perceptions and Behavioral Intention to use AI responsibly.

AI Anxiety, as an affective response independent of cognitive risk evaluation, is expected to dampen the extent to which favourable perceptions of AI translate into behavioral intention. Chen et al. (2024) demonstrate that AI anxiety exerts a direct negative effect on AI use, while

Cao et al. (2026) and Nghĩa (2026) confirm its relevance within integrated TAM–UTAUT frameworks for GenAI adoption. Extending this logic to a moderating role, individuals experiencing higher AI anxiety are expected to convert the same underlying UTAUT-based cognitions into weaker behavioral intention than individuals with lower anxiety. Accordingly:

H6: AI Anxiety moderates the relationship between overall UTAUT-related perceptions and Behavioral Intention to use AI responsibly.

Table 1

Summary of Research Hypotheses

Hypothesis	Statement
H1	Effort Expectancy (EE) has a positive effect on Behavioral Intention (BI) to use AI responsibly.
H2	Performance Expectancy (PE) has a positive effect on Behavioral Intention (BI) to use AI responsibly.
H3	Social Influence (SI) has a positive effect on Behavioral Intention (BI) to use AI responsibly.
H4	Facilitating Conditions (FC) has a positive effect on Behavioral Intention (BI) to use AI responsibly.
H5	Responsible AI Awareness (RA) moderates the relationship between UTAUT predictors (EE, PE, SI, FC) and Behavioral Intention (BI), such that the relationship is stronger among individuals with higher RA.
H6	AI Anxiety (ANX) moderates the relationship between UTAUT predictors (EE, PE, SI, FC) and Behavioral Intention (BI), such that the relationship is weaker among individuals with higher ANX.

2.5 Conceptual Model

Integrating the hypotheses above, the proposed research model specifies four direct paths from the core UTAUT predictors — Effort Expectancy, Performance Expectancy, Social Influence, and Facilitating Conditions — to Behavioral Intention to use AI responsibly (H1–H4), together with two moderating paths through which Responsible AI Awareness (H5) and AI Anxiety (H6)

are hypothesized to condition the strength of these relationships. Figure 1 presents the conceptual model.

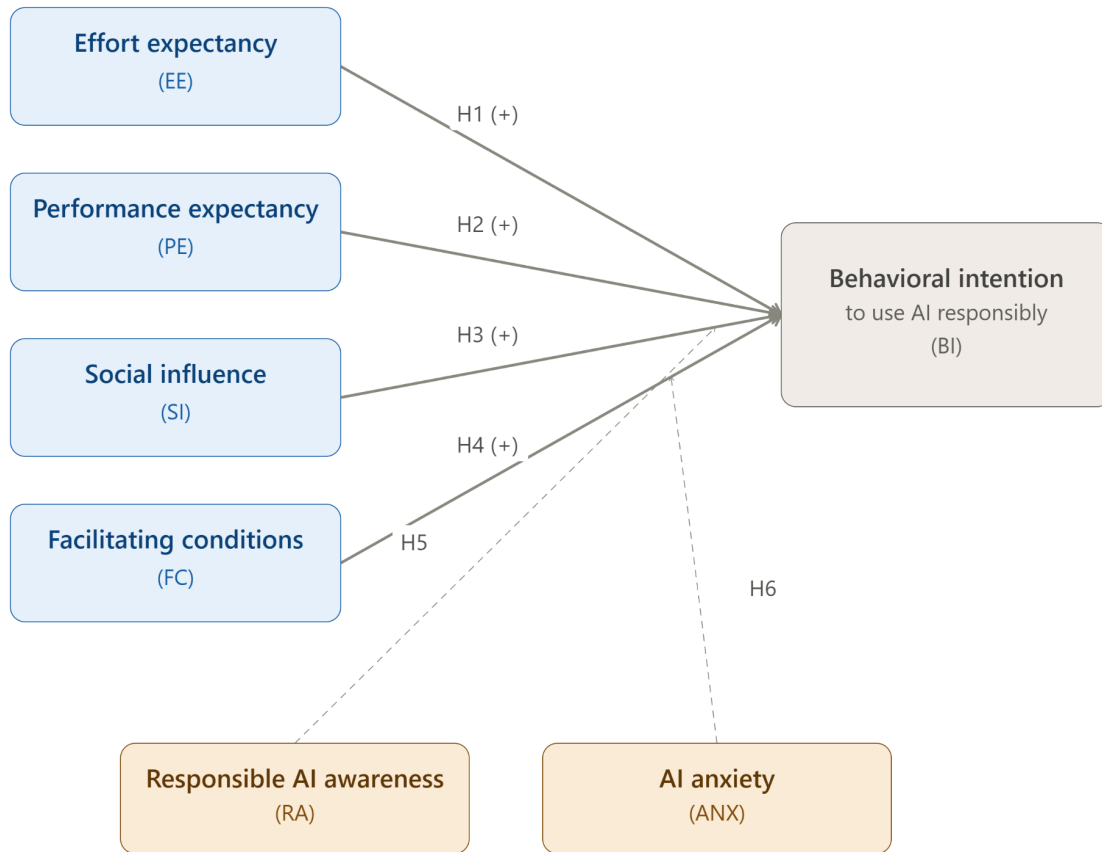


Figure 1. *Conceptual Research Model*

3. Methodology

3.1 Research Design and Sampling

This study adopts a quantitative, cross-sectional research design to examine the factors influencing the behavioral intention of final-year university students and young professionals in Ho Chi Minh City to adopt artificial intelligence (AI) in a responsible manner. A survey-based approach was chosen because it allows for the efficient collection of standardized data from a

relatively large and dispersed population within the study's timeframe, and because it is consistent with prior technology-acceptance research employing structural equation modeling (SEM) (Venkatesh et al., 2012; Kavitha & Joshith, 2024).

The target population consists of two sub-groups that share a common characteristic: they are at the threshold of transitioning from academic training into professional practice. The first sub-group comprises final-year university students who have not yet graduated; the second comprises young professionals who graduated within the past two years. A non-probability convenience and snowball sampling strategy was employed, with respondents recruited through university class groups, alumni networks, professional social-media platforms (LinkedIn), and interest-based community groups on Zalo and Facebook associated with the Ho Chi Minh City Business Association (HUBA). The eligibility criteria required respondents to be (a) either a final-year student not yet graduated, or a graduate with less than two years of professional experience, and (b) based in Ho Chi Minh City.

Following common guidance for covariance- and variance-based SEM, which recommends a minimum ratio of five to ten observations per estimated parameter (Hair et al., 2019), the study set a target sample size of 250 valid responses for a model comprising 35 observed indicators across seven latent constructs. Data collection took place between mid-August and early September 2026, yielding 236 valid responses after data cleaning (see Section 4.1).

3.2 Measurement Instrument

Data were collected using a structured, self-administered online questionnaire distributed via Google Forms, consisting of three sections. Section A captured demographic and screening information. Section B contained the core measurement items, organized into seven constructs, each measured on a five-point Likert scale ranging from 1 ("strongly disagree") to 5 ("strongly agree").

agree"). Section C consisted of two open-ended questions inviting respondents to describe perceived barriers to responsible AI use and their exposure to related training. The seven constructs are Effort Expectancy (EE), Performance Expectancy (PE), Social Influence (SI), and Facilitating Conditions (FC), all adapted from the original UTAUT2 scale (Venkatesh et al., 2012); Responsible AI Awareness (RA), capturing awareness of AI-related risks such as data-privacy concerns and the need for transparency; AI Anxiety (ANX), capturing apprehension about job displacement and over-reliance on AI (adapted from Caffaratti et al., 2025); and Behavioral Intention (BI), the dependent construct. Each construct was measured with five items, yielding 35 observed indicators in total. Table 2 summarizes the constructs.

Table 2

Summary of Measurement Constructs

Construct	Code	Items	Sample item	Source
Effort Expectancy (EE)	EE1–EE5	5	"I find AI tools easy to use without much technical knowledge."	Adapted from Venkatesh et al. (2012)
Performance Expectancy (PE)	PE1–PE5	5	"Using AI helps me complete tasks faster."	Adapted from Venkatesh et al. (2012)
Social Influence (SI)	SI1–SI5	5	"People around me encourage me to use AI."	Adapted from Venkatesh et al. (2012)
Facilitating Conditions (FC)	FC1–FC5	5	"I have the resources necessary to use AI tools."	Adapted from Venkatesh et al. (2012)
Responsible AI Awareness (RA)	RA1–RA5	5	"I am aware of the potential privacy risks associated with AI"	Adapted from ethics-awareness items in AI-literacy research
AI Anxiety (ANX)	ANX1–ANX5	5	"I worry that AI will replace my job in the near future."	Adapted from Caffaratti et al. (2025)

Construct	Code	Items	Sample item	Source
Behavioral Intention (BI)	BI1–BI5	5	“I intend to use AI responsibly in my study or work”	Adapted from Venkatesh et al. (2012)

Prior to full-scale distribution, the questionnaire was piloted with a small convenience sample ($n \approx 30$) to assess item clarity and completion time; minor wording adjustments were made based on pilot feedback. The Vietnamese-language version of the instrument was used for data collection, with all items back-translated to confirm conceptual equivalence with their English-language sources.

3.3 Data Collection Procedure

The survey link, together with a brief description of the study's purpose, estimated completion time (5–7 minutes), and an assurance of anonymity and confidentiality, was distributed through the channels described in Section 3.1. Two data-quality safeguards were applied: straight-lining responses (identical answers across all 35 Likert items) were flagged and removed prior to analysis, as such patterns typically indicate inattentive responding (DeSimone et al., 2015), and records with missing values on core Likert items were excluded case-wise.

3.4 Data Analysis Method

Data analysis proceeded in four sequential stages, consistent with standard practice for survey-based SEM research (Hair et al., 2019). First, descriptive statistics were computed for demographic variables and each observed indicator. Second, internal consistency reliability was assessed using Cronbach's alpha (α), with corrected item-total correlations and "alpha-if-item-deleted" values examined to identify weak items. Third, exploratory factor analysis (EFA) with promax rotation was conducted to examine the underlying factor structure,

evaluating sampling adequacy via the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's test of sphericity.

Fourth, the measurement and structural models were estimated using **covariance-based structural equation modeling (CB-SEM)**, implemented in Python via the *semopy* library (Igolkina & Meshcheryakov, 2020), rather than partial least squares SEM software. CB-SEM was selected because it enables formal confirmatory testing of the measurement model's overall fit (χ^2 , CFI, TLI, RMSEA) prior to structural estimation, and performs reliably at the achieved sample size ($N = 236$) for a model of this complexity (Hair et al., 2019). The measurement model was evaluated for indicator reliability, convergent validity (average variance extracted, $AVE \geq 0.50$), and discriminant validity (Heterotrait-Monotrait ratio, $HTMT < 0.90$; Henseler et al., 2015). Factor scores extracted from the confirmatory factor analysis (CFA) were standardized (z-scored) prior to structural estimation. Interaction terms for the moderation hypotheses (H5 and H6) were computed as the product of mean-centred scores of the moderator (RA or ANX) and a composite UTAUT2 index (the average of standardized EE, PE, SI, and FC scores). Structural path coefficients, standard errors, and 95% confidence intervals were obtained via bootstrapping with 5,000 resamples. All analyses were conducted in Python (pandas, factor_analyzer, semopy, statsmodels); full results are reported in Section 4.

4. Results

4.1 Sample Description and Demographic Profile

A total of 236 valid responses were collected via an online Google Form survey distributed through university networks, alumni groups, and professional social media channels from mid-August to early September 2026. All items were measured on a five-point Likert scale.

Responses were screened for straight-lining prior to analysis; no respondents were removed on this basis in the final dataset, and no missing values were detected across the 35 Likert items.

As presented in Table 3, the sample was composed predominantly of final-year undergraduate students ($n = 177$, 75.0%), with the remaining quarter ($n = 59$, 25.0%) comprising recent graduates employed for fewer than two years. In terms of academic background, the largest group was Technology/IT/Data Science (32.6%), followed by Social Sciences/Humanities (26.3%), Economics/Business (23.3%), and Other fields (17.8%). Regarding prior AI experience, 83.9% of respondents reported frequent use of AI tools (several times per week or more), reflecting a sample already actively engaged with AI.

Table 3

Demographic Profile of Respondents ($N = 236$)

Variable	Category	n	%
Respondent Group	Final-year students (pre-graduation)	177	75.0%
	Recent graduates (< 2 years working)	59	25.0%
Field of Study	Technology / IT / Data Science	77	32.6%
	Social Sciences / Humanities / Linguistics	62	26.3%
	Economics / Business / Accounting	55	23.3%
	Other fields	42	17.8%
AI Usage Frequency	Frequent (several times/week or more)	198	83.9%
	Tried once or twice	31	13.1%
	Never used	7	3.0%

4.2 Reliability and Validity Assessment

Internal consistency was evaluated using Cronbach's alpha (α), composite reliability (CR), and average variance extracted (AVE) for each of the seven constructs (Table 4). All constructs demonstrated satisfactory Cronbach's alpha values, ranging from 0.782 (FC) to 0.876 (PE), with

six of seven constructs exceeding the 0.80 threshold recommended by Hair et al. (2019). CR values ranged from 0.786 to 0.876, all above the 0.70 threshold.

AVE values ranged from 0.426 to 0.586. Five of seven constructs achieved $AVE \geq 0.50$ (Fornell & Larcker, 1981). Two constructs fell marginally below this threshold — EE (AVE = 0.490) and ANX (AVE = 0.494) — but are supported by CR values well above 0.70; FC (AVE = 0.426) represents the most notable deviation. Fornell and Larcker (1981) note that when CR exceeds 0.70 and factor loadings are uniformly significant, construct validity can be considered adequate; FC meets both conditions (CR = 0.786; all loadings significant at $p < .05$), and all HTMT values involving FC fall below 0.85 (maximum 0.675), providing supplementary evidence of discriminant validity. FC was therefore retained in the structural model. Corrected item-total correlations for all 35 items exceeded 0.30 (range: 0.459–0.732), confirming that no items required deletion.

Table 4

Reliability, Composite Reliability, and Average Variance Extracted

Construct	Items	Loading Range	Cronbach's α	CR	AVE	Assessment
EE – Effort Expectancy	5	0.667–0.751	0.827	0.827	0.490	Acceptable*
PE – Performance Expectancy	5	0.739–0.798	0.876	0.876	0.586	Good
SI – Social Influence	5	0.605–0.766	0.840	0.841	0.515	Good
FC – Facilitating Conditions	5	0.525–0.708	0.782	0.786	0.426	Marginal†
RA – Responsible AI Awareness	5	0.550–0.817	0.833	0.842	0.520	Good
ANX – AI Anxiety	5	0.670–0.741	0.829	0.830	0.494	Acceptable*
BI – Behavioral Intention	5	0.683–0.786	0.841	0.844	0.520	Good

* *AVE slightly below 0.50; CR > 0.70 and all loadings significant — convergent validity considered adequate (Fornell & Larcker, 1981).* † *AVE = 0.426; retained based on CR = 0.786, significant loadings, and HTMT < 0.85 for all pairs.*

Discriminant validity was assessed using the Heterotrait-Monotrait ratio (HTMT), recommended by Henseler, Ringle, and Sarstedt (2015) as a superior alternative to the Fornell-Larcker criterion. As shown in Table 5, all HTMT values were below the conservative threshold of 0.85, with the highest value observed between EE and FC (HTMT = 0.675), confirming that all seven constructs are empirically distinct.

Table 5

Heterotrait-Monotrait Ratio (HTMT) Matrix

	EE	PE	SI	FC	RA	ANX	BI
EE	—	0.583	0.420	0.675	0.497	0.438	0.446
PE	0.583	—	0.499	0.552	0.425	0.360	0.393
SI	0.420	0.499	—	0.432	0.496	0.434	0.508
FC	0.675	0.552	0.432	—	0.405	0.395	0.428
RA	0.497	0.425	0.496	0.405	—	0.415	0.431
ANX	0.438	0.360	0.434	0.395	0.415	—	0.378
BI	0.446	0.393	0.508	0.428	0.431	0.378	—

Note. All values < 0.85; discriminant validity confirmed (Henseler et al., 2015).

4.3 Measurement Model Fit

The overall fit of the confirmatory factor analysis (CFA) measurement model was evaluated using multiple indices, following the two-step approach recommended by Anderson and Gerbing (1988). Results are presented in Table 6. The chi-square to degrees-of-freedom ratio ($\chi^2/df = 1.450$) was well within the acceptable range (< 3.0). The incremental fit indices CFI (0.931) and TLI (0.924) both exceeded the 0.90 threshold, and RMSEA (0.044) fell within the 'good fit'

range ($\leq .05$). GFI (0.811) and AGFI (0.791) fell below the conventional 0.90 threshold; these indices are known to be sensitive to model complexity and sample size (Marsh, Balla & McDonald, 1988), and their use has been discouraged in favour of CFI, TLI, and RMSEA in models with a large number of observed variables (Hair et al., 2019). Given that the present model contains 35 observed variables across seven latent constructs, the marginal GFI/AGFI values are consistent with expectations and do not invalidate the model.

Table 6

CFA Model Fit Indices

Index	Value	Threshold	Assessment
χ^2/df	781.46 / 539 = 1.450	< 3.0	Good
CFI	0.931	≥ 0.90	Acceptable
TLI	0.924	≥ 0.90	Acceptable
RMSEA	0.044	≤ 0.08	Good
GFI	0.811	≥ 0.90	Marginal‡
AGFI	0.791	≥ 0.85	Marginal‡

‡ GFI and AGFI are sensitive to model complexity; Hair et al. (2019) recommend CFI, TLI, and RMSEA as primary indices for complex models.

4.4 Structural Model and Hypothesis Testing

The structural model was estimated using CB-SEM via semopy, with bootstrapping (5,000 iterations) applied to obtain robust standard errors and 95% confidence intervals for each path coefficient. Results are presented in Table 7.

Of the six hypotheses tested, two were statistically supported at $\alpha = .05$. H3 received the strongest support: Social Influence exhibited the largest and most significant effect on Behavioral Intention ($\beta = 0.379$, $t = 4.619$, $p < .001$, 95% CI [0.214, 0.536]). H1 was also

supported, with Effort Expectancy demonstrating a positive and significant effect on BI ($\beta = 0.211$, $t = 2.025$, $p = .044$).

Four hypotheses were not supported. The non-significance of Performance Expectancy (H2: $\beta = -0.006$, $p = .938$) is a theoretically notable finding, given that PE is typically among the strongest predictors in the UTAUT/TAM literature; this pattern is consistent with recent findings by Kavitha and Joshith (2024), who observed attenuated PE effects in digital-native cohorts with high prior technology exposure. Facilitating Conditions (H4: $\beta = 0.066$, $p = .434$) did not significantly predict BI, consistent with the 83.9% frequent-AI-use rate observed in the sample. The moderation hypotheses H5 ($RA \times UTAUT2$; $\beta = 0.052$, $p = .453$) and H6 ($ANX \times UTAUT2$; $\beta = 0.078$, $p = .296$) were not supported, which may reflect the statistical power required to detect interaction effects (Aguinis, 2004) and the compression of variance introduced by the composite UTAUT2 index used as the focal predictor for moderation.

Table 7

Structural Model Results: Hypothesis Testing (Bootstrap $n = 5,000$)

Hyp.	Path	β	SE	t	p	95% CI	Result
H1	EE $\rightarrow$ BI	0.211	0.104	2.025	0.044	[0.004, 0.415]	Supported
H2	PE $\rightarrow$ BI	-0.006	0.079	-0.071	0.938	[-0.161, 0.149]	Not supported
H3	SI $\rightarrow$ BI	0.379	0.082	4.619	< .001	[0.214, 0.536]	Supported
H4	FC $\rightarrow$ BI	0.066	0.085	0.785	0.434	[-0.099, 0.232]	Not supported
H5	$RA \times UTAUT2 \rightarrow$ BI (mod.)	0.052	0.072	0.721	0.453	[-0.097, 0.182]	Not supported
H6	$ANX \times UTAUT2 \rightarrow$ BI (mod.)	0.078	0.079	0.993	0.296	[-0.092, 0.226]	Not supported

Note. β = standardized path coefficient (bootstrap mean); SE = bootstrap standard error; CI = 95% bias-corrected confidence interval; mod. = moderation term (product of mean-centred scores).

In summary, the structural model explains behavioral intention to adopt AI responsibly primarily through social normative pressure (SI) and perceived ease of use (EE). The model explained 45.4% of the variance in Behavioral Intention, indicating moderate explanatory power.

5. Discussion and Conclusion

5.1 Discussion of Findings

This study examined the factors influencing behavioural intention to adopt artificial intelligence (AI) responsibly among final-year students and young professionals in Ho Chi Minh City, integrating the core predictors of UTAUT2 (Venkatesh et al., 2012) with two context-specific constructs — Responsible AI Awareness (RA) and AI Anxiety (ANX). Of the six hypotheses tested, two received statistical support, and the pattern of significant and non-significant paths offers a coherent, theoretically interpretable account of AI readiness in a digital-native population.

The most striking finding is the dominance of **Social Influence** as a predictor of behavioural intention (H3: $\beta = 0.379$, $p < .001$), by a considerable margin the strongest effect in the model. This suggests that for this cohort, the decision to adopt AI responsibly is fundamentally a social one, shaped by peers, mentors, and prevailing professional norms rather than by individual calculations of usefulness. This finding departs from much of the earlier UTAUT literature, in which Performance Expectancy is typically the dominant predictor of technology acceptance (Venkatesh et al., 2012), and instead resonates with more recent work on Gen Z technology adoption (Kavitha & Joshith, 2024).

Effort Expectancy also emerged as a significant, though comparatively modest, predictor (H1: $\beta = 0.211$, $p = .044$), indicating that perceived ease of use retains explanatory value even among

frequent users (83.9% of the sample). This may reflect a distinction between ease of *basic* use, which is nearly universal, and ease of applying AI *responsibly* — for example, verifying outputs or managing data privacy — which may still demand effort not fully commoditised by user-friendly interfaces.

By contrast, the non-significance of **Performance Expectancy** (H2: $\beta = -0.006$, $p = .938$) constitutes a theoretically important null result. Given that PE is conventionally the strongest predictor in technology-acceptance research (Venkatesh et al., 2012), its complete absence of effect here suggests a ceiling phenomenon: for a population in which the usefulness of AI is already taken for granted, PE loses its power to differentiate between higher and lower intenders. Similarly, **Facilitating Conditions** did not significantly predict behavioural intention (H4: $\beta = 0.066$, $p = .434$), plausibly because infrastructural access to AI tools is no longer a meaningful constraint for an urban, university-affiliated population in Ho Chi Minh City.

The two moderation hypotheses were not supported: neither **Responsible AI Awareness** (H5: $\beta = 0.052$, $p = .453$) nor **AI Anxiety** (H6: $\beta = 0.078$, $p = .296$) significantly moderated the relationship between the composite UTAUT2 predictors and behavioural intention. Three explanations merit consideration. First, moderation effects typically require substantially larger samples than main-effect tests to achieve adequate statistical power (Aguinis, 2004); the present sample of 236 may be underpowered for interaction effects. Second, aggregating the four UTAUT into a single composite index may have compressed meaningful variance, since RA or ANX could plausibly moderate some paths more strongly than others. Third, among a sample already characterised by frequent AI use, both awareness of AI's risks and anxiety about its consequences may have become normalised background conditions rather than active moderators of behaviour.

Taken as a whole, the structural model explained 45.4% of the variance in behavioural intention ($R^2 = .454$), exceeding the 0.25 threshold recommended for substantial explanatory power (Hair et al., 2019).

5.2 Theoretical Contributions

This study contributes to the technology-acceptance literature in three ways. First, it extends UTAUT2 into the specific normative domain of responsible AI use by explicitly incorporating Responsible AI Awareness as a construct in its own right; while its moderating role was not confirmed, its inclusion as a distinct, reliably measured construct ($\alpha = .833$, AVE = .520) opens a path for future models to test RA as a direct antecedent rather than solely as a moderator. Second, the finding that Social Influence overwhelmingly outperforms Performance Expectancy challenges the assumed universality of UTAUT2's predictor hierarchy and adds to evidence that generational and contextual factors reorder the relative importance of technology-acceptance predictors (Kavitha & Joshith, 2024). Third, the non-significant moderation results for both RA and ANX are themselves a contribution, indicating that further theoretical and methodological refinement is needed before firm conclusions can be drawn about how awareness and anxiety shape responsible AI adoption.

5.3 Practical Implications

For higher education institutions, the dominance of Social Influence suggests that interventions built around peer modelling and visible institutional endorsement are likely to be more effective at cultivating responsible AI adoption than technical training focused solely on tool proficiency, which the near-universal frequent-use rate (83.9%) suggests is already largely achieved. For employers, onboarding processes emphasising visible managerial and peer expectations around

responsible AI use may shift behaviour more effectively than emphasising productivity gains, which appear already assumed rather than persuasive. For policymakers, findings suggest that public awareness campaigns relying primarily on communicating AI's usefulness may have limited marginal effect on this population; campaigns leveraging visible social proof may be a more effective lever for shifting responsible-use behaviour at scale.

5.4 Limitations and Future Research

Several limitations qualify these findings. First, the cross-sectional design precludes causal inference. Second, the non-probability convenience and snowball sampling strategy limits generalisability beyond Ho Chi Minh City and may over-represent individuals already engaged with AI. Third, all measures relied on self-report, introducing possible common-method and social-desirability bias. Fourth, future studies should test RA and ANX as moderators of each UTAUT2 predictor individually rather than of a single composite index, which may reveal moderation effects concentrated on specific paths. Finally, future research could extend this model by testing behavioural intention against actual observed responsible-AI behaviour, addressing the well-documented intention-behaviour gap in technology-acceptance research. The relatively low AVE of Facilitating Conditions should also be acknowledged as a measurement limitation and suggests that the construct may require refinement in future research.

5.5 Conclusion

This study examined what drives final-year students and young professionals in Ho Chi Minh City to form intentions to use artificial intelligence responsibly, integrating UTAUT2 with two purpose-built constructs addressing awareness of AI's risks and anxiety about its consequences. Behavioural intention in this population is shaped primarily by social influence and, to a lesser

extent, perceived ease of use, while perceived usefulness and facilitating conditions no longer differentiate intention in a cohort for whom AI has become a taken-for-granted tool. The hypothesised moderating roles of Responsible AI Awareness and AI Anxiety were not confirmed, pointing to the need for larger, more granular follow-up studies. As AI adoption approaches saturation among young, urban, digitally native populations, the locus of influence over how AI is used responsibly is shifting from individual utilitarian calculation toward the visible norms of one's social and professional environment. Educational institutions, employers, and policymakers seeking to promote responsible AI use would therefore do well to invest as much in shaping visible social example as in expanding technical access or communicating AI's benefits.

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